DA07 A table in Python: DataFrame, columns and reading a CSV

Week 3, Tables in pandas | Session 7 of 24 | Python for Data Analysis | New library: pandas

Today's question. What is in each table we have been given?

Objective. By the end: you can build a small table, read a CSV file into one, and say how many rows and columns it has and what each column holds.

Before this session: DA06 Functions and checks that fail loudly. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
row	One record in a table, such as one sale.	One line in the shop register.
column	One kind of information in a table, such as units.	One vertical strip of the register.
header	The name written at the top of a column.	The title of a column in the register.
CSV	Comma-separated values: a plain-text file that holds a table.	A register copied by hand with commas between the boxes.
DataFrame	The table object in pandas, with rows and named columns.	The register after it has been opened in Python.
library	Code written by other people that you load with import.	A toolbox you borrow.

Block A. Build a tiny table

pandas is a library for tables. import pandas as pd loads it and gives it the short name pd. A dot after a name reaches something that belongs to it: sales.shape asks the table for its size. A shape of (3, 2) means three rows and two columns.

Predict. Predict the shape of the table before you run the cell. Write your guess on paper first.

Your guess, written before you run it

```
import pandas as pd
sales = pd.DataFrame({"item": ["Book", "Pen", "Book"], "units": [2, 5, 3]})
print(sales)
print(sales.shape)
print(list(sales.columns))
```

Run it, then compare. The cell prints a table of three rows and two columns, then (3, 2), then ['item', 'units']. The first column holds the item sold and the second holds the units.

Investigate.

1. Which number in (3, 2) counts rows?
2. Where do the column headers come from?

Modify. Add a third column called price_ugx with the values 3000, 1000 and 3000, then print the shape.

Block B. Read a CSV file

pd.read_csv reads a CSV file and returns a DataFrame. The path is where the file is. Here the file sits beside the notebook, so its name is enough. dtypes lists the type of each column.

Predict. Predict how many rows and columns camp_shop.csv has, and which columns hold text. Write your guess on paper first.

Your guess, written before you run it

```
shop = pd.read_csv("camp_shop.csv")
print(shop.shape)
print(shop.dtypes)
products = pd.read_csv("sunrise_products.csv")
print(products.shape)
print(products.head(3))
```

Run it, then compare. The shop table has shape (60, 5). date and item hold text. The other three columns are whole numbers. The Sunrise products table has shape (15, 5): fifteen products, each with an id, a name, a category and two prices.

Investigate.

1. Which columns would you use to work out revenue?
2. Why is date a text column here?

Modify. Show the last three rows of the shop table with tail(3).

Make

Read sunrise_sales.csv, the Sunrise Shops sales. Print its shape and its column names. Write which column holds the date and which holds the price of one unit.

Save your work as DA07. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
FileNotFoundError: [Errno 2] No such file: 'sunrise_sales.csv'	Python cannot find the file where it looks.	Put the notebook and the CSV in the same folder, and check the spelling of the file name.
ModuleNotFoundError: No module named 'pandas'	This laptop is not using the prepared environment.	Ask the instructor. Do not install random packages.

Exit ticket

A DataFrame is a table with rows and named columns, and read_csv turns a CSV file into one.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or assessments. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been submitted until you use the route in START_HERE.pdf.